

Comparative Risk of Hyperglycemia with PI3K, AKT, and mTOR Inhibitors in Breast Cancer



Shreya Ghanshyam Patel¹, Vasu Malhotra¹, Momna Nisar², Mai M Abouzeid³, Ashish Nepal⁴, Aditya Khanijo⁵, Lavanya Nagarajan⁶, Muhammad Zain Farooq¹, Anas Bizanti¹, Pooja Prem Advani⁷

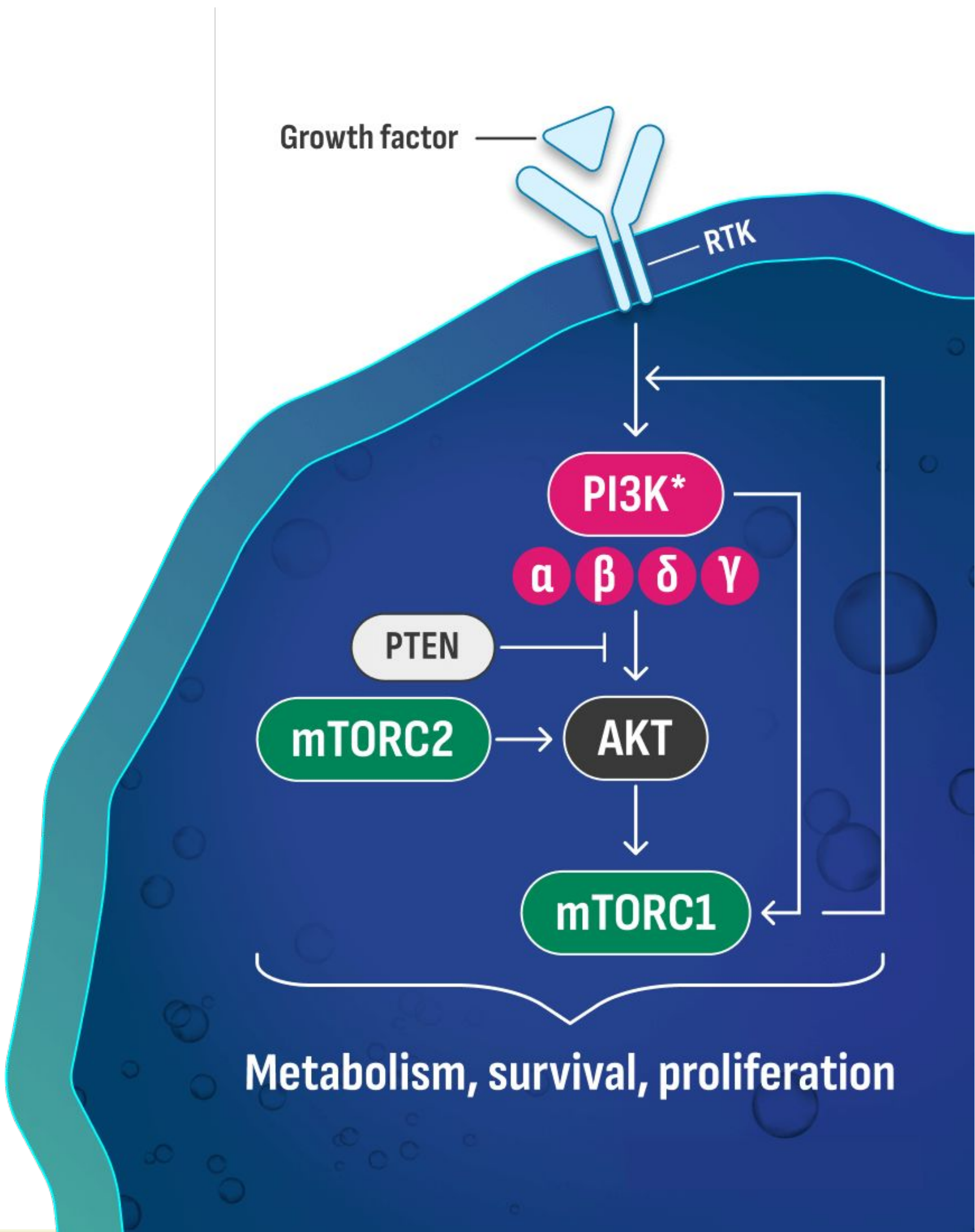
¹Lakeland Regional Health, Lakeland, FL ²Woodhull Medical Center, Brooklyn, NY ³North Mississippi Medical Center, Tupelo, MS ⁴WellSpan York Hospital, York, PA ⁵Saint Vincent Hospital, Worcester, MA ⁶Rosalind Franklin University of Medicine and Science, North Chicago, IL ⁷Department of Medical Oncology, Mayo Clinic, Jacksonville, FL

INTRODUCTION

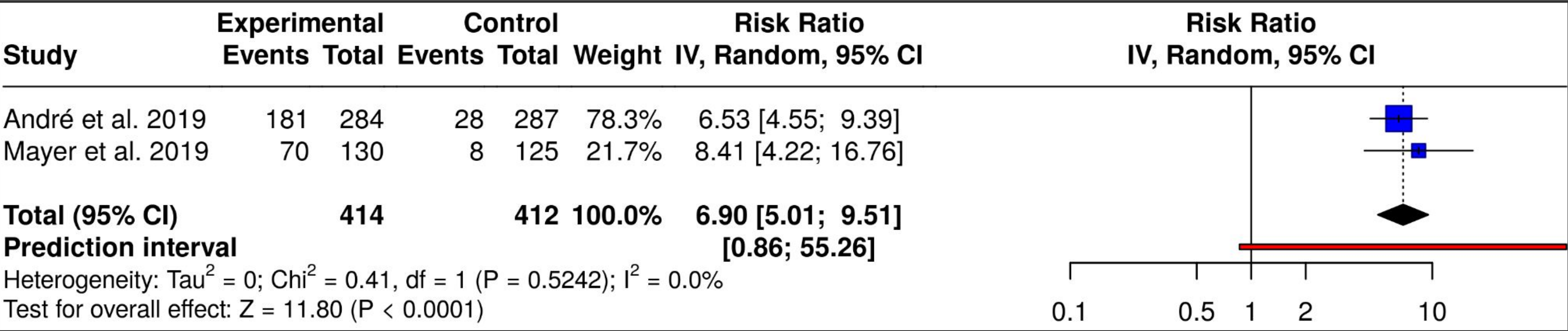
Targeting the PI3K-AKT-mTOR (PAM) signaling pathway has become a central therapeutic strategy in advanced breast cancer, particularly in hormone receptor-positive disease. Multiple agents across this pathway are now in clinical use or late-stage development, including the PI3K inhibitors alpelisib and inavolisib, the AKT inhibitor capivasertib, and the mTOR inhibitor everolimus. Although these agents enhance disease control, pathway inhibition can impair insulin signaling and commonly cause treatment-related hyperglycemia, sometimes requiring dose interruption, discontinuation, or metabolic management. Despite the growing use of these therapies, the comparative risk of hyperglycemia across these agents is not well defined.

METHODS

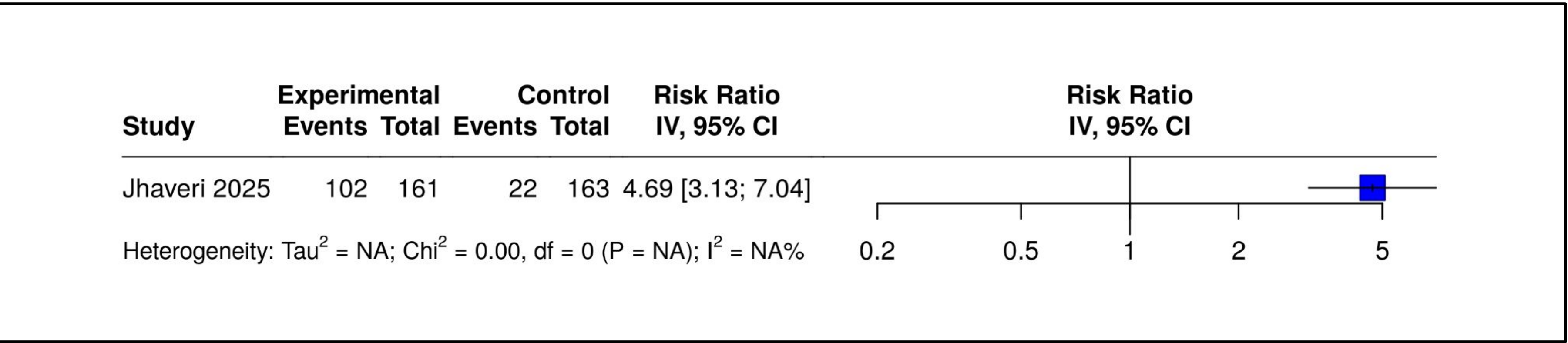
A systematic literature search was performed across PubMed/MEDLINE, Embase (OVID), CENTRAL, ClinicalTrials.gov, and Web of Science from database inception through January 1, 2025. A total of 4,104 records were identified and screened. Eligible studies were randomized controlled trials (RCTs) in breast cancer comparing a PAM pathway inhibitor at a standard dose versus a control regimen lacking that agent. Standard doses included for the analysis were alpelisib 300 mg, inavolisib 9 mg, taselisib 4 mg, capivasertib 400 mg twice daily, and everolimus 10 mg. Trials reporting any-grade hyperglycemia were included.



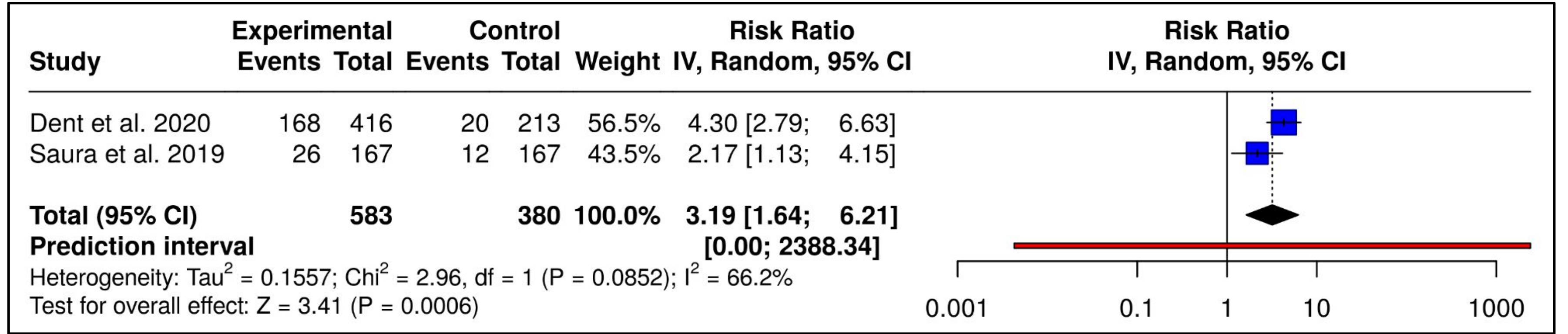
Alpelisib



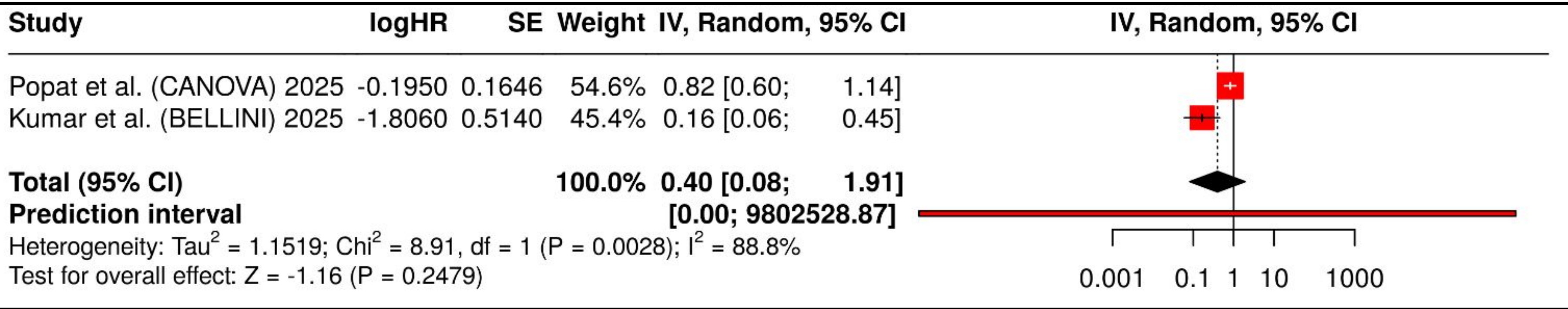
Inavolisib



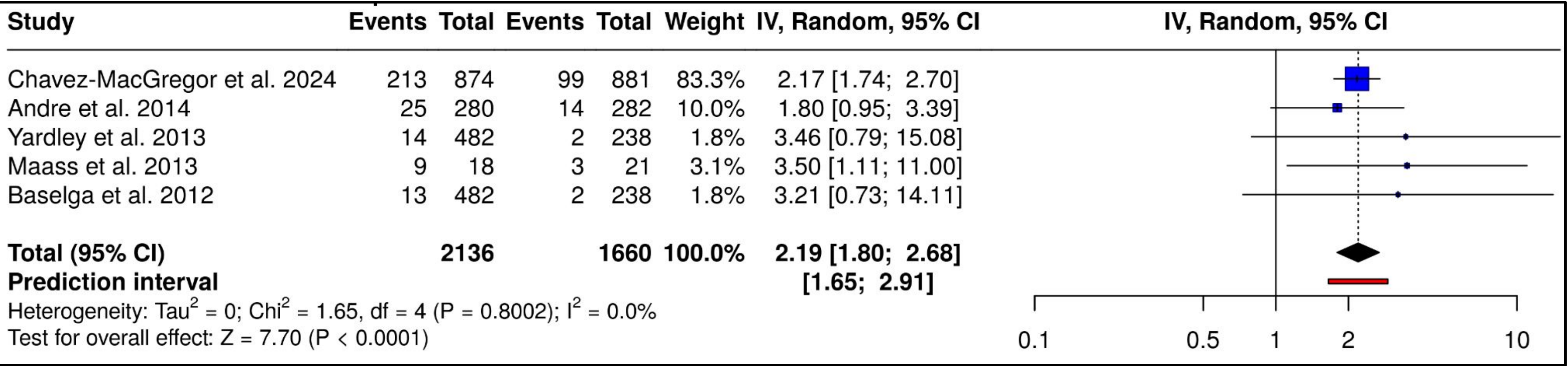
Taselisib



Capivasertib



Everolimus



RESULTS

Across 14 randomized trials including 6,940 patients, all PAM pathway inhibitors were associated with increased hyperglycemia risk compared with control regimens. Alpelisib demonstrated the highest risk of hyperglycemia (any-grade RR 6.90, 95% CI 5.01–9.51; grade ≥ 3 RR 58.75, 95% CI 16.96–203.49). Inavolisib was also associated with increased risk (any-grade RR 4.69, 95% CI 3.13–7.04; grade ≥ 3 RR 23.28, 95% CI 1.38–391.84; single trial). Capivasertib and taselisib showed intermediate risk (capivasertib: any-grade RR 3.53, 95% CI 2.42–5.16; grade ≥ 3 RR 10.02, 95% CI 2.70–37.16; taselisib: any-grade RR 3.19, 95% CI 1.64–6.21; grade ≥ 3 RR 14.61, 95% CI 2.79–76.38). Everolimus demonstrated the lowest relative risk (any-grade RR 2.19, 95% CI 1.80–2.68), with grade ≥ 3 hyperglycemia not reaching statistical significance (RR 5.89, 95% CI 0.74–47.13).

CONCLUSIONS

All PAM pathway inhibitors were associated with an increased risk of hyperglycemia compared with control regimens, with clear stratification by drug class. PI3K inhibitors demonstrated the highest relative risk for both any-grade and grade ≥ 3 hyperglycemia, AKT inhibitors showed an intermediate risk, and mTOR inhibitors the lowest risk. This severity-dependent gradient likely reflects increasing proximity of pathway inhibition to insulin receptor signaling, with PI3K α inhibition most directly disrupting glucose homeostasis. Interpretation is limited by heterogeneity in trial design and limited randomized data for newer agents.

CLINICAL IMPORTANCE:

PAM inhibitors are a common, under-recognized cause of severe hyperglycemia in cancer patients

Risk varies markedly by drug class: PI3K inhibitors carry the highest risk, followed by AKT inhibitors, with lower risk from mTOR inhibitors.

Baseline A1c and early glucose monitoring are critical when these agents are initiated.

Severe (grade ≥ 3) hyperglycemia is not rare and can require insulin, hospitalization, or treatment interruption.

Figure 1 Summary of Pooled Outcomes for any-grade hyperglycemia